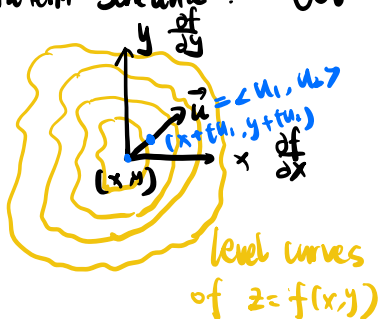


Lecture 7 Directional Derivatives

Midterm Schedule: Oct 19th, Sunday, 19:00 - 21:30, E4-102



Def (Directional Derivatives). Given a unit direction $\vec{u} = u_1 \vec{i} + u_2 \vec{j}$ and a two-variable function $f(x, y)$, the directional derivative of f in the direction of $\vec{u}$ at point (x, y) is defined to be

$$D_{\vec{u}} f(x, y) = \left. \frac{d}{dt} f(x + t u_1, y + t u_2) \right|_{t=0}$$
$$= \lim_{t \rightarrow 0} \frac{f(x + t u_1, y + t u_2) - f(x, y)}{t}$$

When $\vec{u} = \vec{i}$, $D_{\vec{i}} f(x, y) = \left. \frac{d}{dt} f(x + t, y) \right|_{t=0} = \lim_{t \rightarrow 0} \frac{f(x + t, y) - f(x, y)}{t} = \frac{\partial f(x, y)}{\partial x}$
" $\langle 1, 0 \rangle$

Def. (Gradient Vector). Given $f(x, y) \in C^1$, the gradient vector of $f(x, y)$ is defined as $\nabla f(x, y) = \frac{\partial f}{\partial x}(x, y) \vec{i} + \frac{\partial f}{\partial y}(x, y) \vec{j}$.

Example. $f(x, y) = x^2 y + x^3$

$$\frac{\partial f}{\partial x} = 2xy + 3x^2 \quad \frac{\partial f}{\partial y} = x^2$$

$$\nabla f(x, y) = (2xy + 3x^2) \vec{i} + x^2 \vec{j}$$

$$\nabla f(1, 1) = 5 \vec{i} + \vec{j}$$

Thm. Given $f(x,y) \in C^1$, the directional derivative of f at (x,y) in the unit direction $\vec{u} = u_1 \vec{i} + u_2 \vec{j}$ is given by

$$D_{\vec{u}} f(x,y) = \nabla f(x,y) \cdot \vec{u}$$

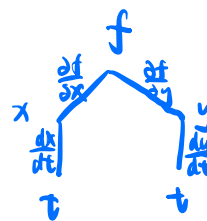
Special case: $\vec{u} = \vec{i}$

$$D_{\vec{i}} f(x,y) = \left(\frac{\partial f}{\partial x}(x,y) \vec{i} + \frac{\partial f}{\partial y}(x,y) \vec{j} \right) \cdot \vec{i} = \frac{\partial f}{\partial x}(x,y)$$

Pf. $D_{\vec{u}} f(x_0, y_0)$ along $\vec{r}(t) = \langle x_0, y_0 \rangle + t \langle u_1, u_2 \rangle$

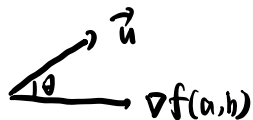
$$\begin{cases} x = x_0 + t u_1 \\ y = y_0 + t u_2 \end{cases}$$

$$D_{\vec{u}} f(x_0, y_0) = \frac{d}{dt} \underbrace{f(x_0 + t u_1, y_0 + t u_2)}_{\text{single variable function of } t} \Big|_{t=0}$$



$$\begin{aligned} D_{\vec{u}} f &= \frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} \\ &= \frac{\partial f}{\partial x} \frac{d(x_0 + t u_1)}{dt} + \frac{\partial f}{\partial y} \frac{d(y_0 + t u_2)}{dt} \\ &= \frac{\partial f}{\partial x} \cdot u_1 + \frac{\partial f}{\partial y} \cdot u_2 \\ &= \left(\frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} \right) \cdot (u_1 \vec{i} + u_2 \vec{j}) \\ &= \nabla f \cdot \vec{u} \end{aligned}$$

$$D_{\vec{u}} f(a,b) = \nabla f(a,b) \cdot \vec{u} = |\nabla f(a,b)| |\vec{u}| \cos \theta = \underline{|\nabla f(a,b)| \cos \theta}$$



$D_{\vec{u}} f(a,b)$ is maximized when $\cos \theta = 1$, i.e. $\theta = 0$, $\nabla f(a,b) \parallel \vec{u}$

$\Rightarrow \nabla f(a,b)$ is pointing in the direction at which f increases most rapidly from (a,b)

$\nabla f \perp$ level curve

Thm. Let $f(x,y) \in C^1$, and (a,b) be a point on the level curve $f(x,y)=c$.

Then the gradient vector $\nabla f(a,b)$ is orthogonal to the level curve $f(x,y)=c$ at the point (a,b) .



Pf. Let $\vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}$ be a parametrization of the level curve $f(x,y)=c$

$$f(x(t), y(t)) = c \text{ for any } t$$

$$\frac{d}{dt} f(x(t), y(t)) = 0$$

$$\frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = 0$$

$$\left(\frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} \right) \cdot \left(\frac{dx}{dt} \vec{i} + \frac{dy}{dt} \vec{j} \right) = 0$$

$$\nabla f \cdot \vec{r}'(t) = 0$$

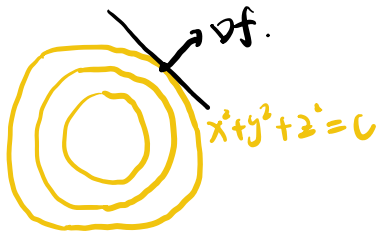
$$\Rightarrow \nabla f \perp \vec{r}'(t) \quad \square$$

$$f(x,y,z) \quad \nabla f = \frac{\partial f}{\partial x} \vec{i} + \frac{\partial f}{\partial y} \vec{j} + \frac{\partial f}{\partial z} \vec{k}$$

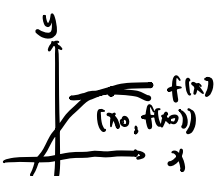
$$\text{unit vector } \vec{u} \quad D_{\vec{u}} f = \nabla f \cdot \vec{u}$$

$\nabla f \perp$ level surface $f(x,y,z)=c$

$f(x,y,z) = x^2 + y^2 + z^2$ level surface $\rightarrow$ sphere



Tangent Planes

$y = f(x)$  $y = f(x_0) + f'(x_0)(x - x_0)$ tangent line

$z = f(x, y)$ tangent plane at $(x_0, y_0, f(x_0, y_0))$

$g(x, y, z) = c$ tangent plane at (x_0, y_0, z_0)

$\begin{cases} \text{any point on the plane } (x_0, y_0, z_0) \\ \text{a normal vector to the plane } \langle A, B, C \rangle \end{cases} \Rightarrow Ax + By + Cz = Ax_0 + By_0 + Cz_0$

Example. Find the tangent plane to the surface $x^2 + y^2 = z^2 + 3$ at the point $(x, y, z) = (2, 0, -1)$

Sol. Let $g(x, y, z) = x^2 + y^2 - z^2$, then the given surface is a level set given by $g(x, y, z) = 3$

$$\nabla g(x, y, z) = \frac{\partial g}{\partial x} \vec{i} + \frac{\partial g}{\partial y} \vec{j} + \frac{\partial g}{\partial z} \vec{k} \\ = 2x \vec{i} + 2y \vec{j} - 2z \vec{k}$$

$$\nabla g(2, 0, -1) = 4 \vec{i} + 2 \vec{k}$$

$$\vec{n} = \langle 4, 0, 2 \rangle$$

$\Rightarrow$ Tangent plane at $(2, 0, -1)$ is given by

$$4x + 0y + 2z = 4(2) + 0(0) + 2(-1)$$

$$4x + 2z = 6$$

Example. Given $f(x, y) = x \cos y - y e^x$, find the tangent plane at $(0, 0, 0)$

Sol. Def $g(x, y, z) = z - x \cos y + y e^x$

$$\nabla g(x, y, z) = (-\cos y + ye^x) \vec{i} + (x \sin y + e^x) \vec{j} + \vec{k}$$

$$\vec{n} = \nabla g(0, 0, 0) = -\vec{i} + \vec{j} + \vec{k} = \langle -1, 1, 1 \rangle$$

$$\Rightarrow \text{tangent plane: } -x + y + z = 0$$

Thm. Given a function $f(x, y) \in C^1$. The equation of the tangent plane for the graph $z = f(x, y)$ at the point $(x_0, y_0, f(x_0, y_0))$ is given by

$$z = f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0).$$

$$g(x, y, z) = z - f(x, y)$$

$$\vec{n} = \nabla g = \left\langle -\frac{\partial f}{\partial x}, -\frac{\partial f}{\partial y}, 1 \right\rangle.$$